

# Hadi Shokati

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## Summary

Soil Scientist and Geospatial Data Scientist with expertise in **remote sensing, UAV photogrammetry, and deep learning** for environmental monitoring. Experienced in building production-ready models (**image segmentation, erosion modelling and forecasting, soil moisture estimation, flood mapping**) and delivering actionable maps and dashboards for agricultural and watershed management. Proficient in **Python, TensorFlow/PyTorch, Google Earth Engine, and GIS workflows**.

## Experience

- Soil Scientist, Department of Geosciences, University of Tübingen, Germany

2023–Present

  - Implement transfer-learning segmentation pipeline for flood and erosion detection, reducing labeled-data requirements by 80% and increasing event detection precision by up to 55%.
  - Develop hybrid models for soil erosion prediction, addressing validation data scarcity challenge in soil erosion studies.
  - Forecast future soil erosion susceptibility across space and time.
  - Reconstruct historical rainfall-erosivity maps, overcoming the lack of high-resolution precipitation records.
  - Automate large-scale data pipelines, reducing preparation time by 40%.
  - Collaborate with computer-science researchers to develop advanced deep-learning models.
  - Advise MSc students on research design, data analysis, and model development.
- Data Scientist, Department of Geosciences, University of Tübingen, Germany

2023

  - Implemented an ML pipeline with UAV hyperspectral data for soil moisture prediction, yielding 33% and 40% R<sup>2</sup> improvements over Landsat and Sentinel-2, respectively.
  - Performed geospatial analysis using GIS, remote, and proximal sensing techniques.
- Geospatial Data Analyst, Soil Conservation and Watershed Management Research Institute, Iran

2022

  - Collected and analyzed UAV data for precision agriculture and watershed management.
  - Integrated remote sensing and UAV data to identify freshwater springs in the Persian Gulf.
- Teaching Assistant, University of Tehran, Iran

2020–2022

  - Courses: Fluid Mechanics, designing surface irrigation systems, Designing drainage and irrigation networks, Surface irrigation hydraulics.

## Technical Skills

- Programming:

Python (Professional), MATLAB (Intermediate), R (Beginner), JavaScript (Beginner)
- GIS and Remote Sensing:

ArcGIS Pro, QGIS, ENVI, SNAP, SAGA and Google Earth Engine (Professional)
- Machine Learning:

Regression, Classification, Data Analysis, Image Processing, Image Segmentation, Transfer Learning, Fine-Tuning, TensorFlow/Keras, PyTorch
- Languages:

English (B2), German (A2), Turkish (B1), Persian and Azerbaijani (C2)
- Other:

UAV Piloting (Flight Operations, Photogrammetry, Spatial Data Analysis)

## Education

- PhD, Soil Science and Geomorphology, University of Tübingen, Germany

2023–Present
- MSc, Water Engineering and Sciences, Tarbiat Modares University, Iran

2019
- BSc, Water Engineering and Sciences, University of Mohaghegh Ardabili, Iran

2017

## Certifications & Training

- Deep Learning certificate, ELLIS Summer School: AI for Earth and Climate Sciences, Germany

2025
- Deep Learning certificate, DeepLearn 2025: 12th International School on Deep Learning, Portugal

2025
- 13th EARSel Workshop on Imaging Spectroscopy, Spain

2024
- German Japanese exchange workshop, Germany

2023